- Sample space → collection of all possible outcomes
- $(A \cup B)^c = A^c \cap B^c$ · $(A \cap B)^c = A^c \cup B^c$ · $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ · $A = (A \cap B) \cup (A \cap B^c)$
- Axioms: $P(A) \geq 0$; $P(S) = 1$; infinite sequence of disjoint events $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$
- $P(\emptyset) = 0$ · $P(A^c) = 1 - P(A)$ · $P(A \cup B) + P(A \cap B) = P(A) + P(B)$
- $P(\bigcup_{i=1}^{n} A_i) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k)$
  $+ \cdots (-1)^{n+1} P(A_1 \cap A_2 \cdots A_n)$
- $P(A|B) = \frac{P(A \cap B)}{P(B)}$ · Independent events $\Rightarrow P(A \cap B) = P(A) \cdot P(B)$
- $A, B \rightarrow$ independent $\Rightarrow A, B^c \rightarrow$ independent
- Mutual independent: $P(A_1 \cap \cdots \cap A_n) = P(A_1) \cdots P(A_n)$
- Pairwise independent: $P(A \cap B) = P(A) \cdot P(B)$; $P(A \cap C) = P(A) \cdot P(C)$;
  $P(B \cap C) = P(B) \, P(C)$; $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$ pairwise & mutual
- $P(A \cap B / C) = P(A|C) \cdot P(B|C) \rightarrow A \& B$ are conditionally ind. given C
- Bayes: $B_1, B_2, \cdots B_K \Rightarrow$ partitions of S $(P(B_j) > 0)$

$$P(B_j|A) = \frac{P(B_j) \cdot P(A|B_j)}{\sum_{j=1}^{K} P(B_j) \cdot P(A|B_j)}$$

- Random variable: real-valued function defined on a sample space S
- Probability function: $f(x) = P(X = x)$; $\sum_x f(x) = 1$
- Support: set of all x such that $f(x) > 0$
- P.D.F: $P(a \leq x \leq b) = \int_a^b f(x)\,dx$; $\int_{-\infty}^{\infty} f(x)\,dx = 1$
- C.D.F: $F(x) = P(X \leq x)$; $\lim_{y \to x^+} F(y) = F(x)$
- $P(x_1 < x \leq x_2) = F(x_2) - F(x_1)$ · $P(X = x) = F(x) - F(x^-)$
- $F(x) = \int_{-\infty}^{x} f(t)\,dt$ · $f(x) = dF(x)/dx$
- Quantile function: for $0 < p < 1$, p quantile x_p is

$F^{-1}(p)$ is smallest x such that $F(x) \geq p$

- $F(x,y) = P(X \leq x \text{ and } Y \leq y)$ • $f(x,y) = P(X = x \text{ and } Y = y)$
- $f(x,y) = \frac{d^2 F(x,y)}{dx\,dy}$
- marginal CDF: $F_1(x) = P(X \leq x) = \lim_{y \to \infty} F(x,y)$
- marginal PF: $f_1(x) = \sum_y f(x,y)$
- marginal PDF: $f_1(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$
- Independent RV's: $P(x \in A \text{ and } y \in B) = P(x \in A) \cdot P(y \in B)$ or $F(x,y) = F_1(x) F_2(y)$ or $f(x,y) = f_1(x) f_2(y)$
- Conditional p.d.f of Y given $X = x$: $g_2(y|x) = f(x,y) / f_1(x)$
- $f(x,y) = g_2(y|x) f_1(x) = g_1(x|y) \cdot f_2(y)$
- $F(x_1, \ldots x_n) = P(x_1 \leq x_1, x_2 \leq x_2 \ldots)$ $f_1(x_1) = \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} f(x_1 \ldots x_n)\,dx_2 \ldots dx_n$
- i.i.d → independent & identically distributed
- Conditional pdf of $(x_1 \ldots x_k)$ given $(x_{k+1} \ldots x_n) = (x_{k+1} \ldots x_n)$ is $f(x_1 \ldots x_n) / f_2(x_{k+1} \ldots x_n)$ if $f_2 \to$ marginal PDF of $(x_{k+1} \ldots x_n)$
- $Y = r(x)$ • discrete: P.f of Y $g(y) = P(Y = y) = \sum_{x : r(x) = y} f(x)$
  - CDF method: $G(y) = P(Y \leq y) = P(r(x) \leq y)$ (calculate by integrating PDF $f(x)$ over set $x : r(x) \leq y$) • $g(y) = G'(y)$
  - Transformation method: If $Y = r(x)$ is one-one, then $X = s(y) \cdot g(y) = f(s(y)) \left| \frac{ds(y)}{dy} \right|$
- If $X \to$ continuous CDF $\Rightarrow Y = F(X) \to \text{Unif}[0,1]$
- $Y_1 = r_1(X_1 \ldots X_n) \ldots Y_m = r_m(X_1 \ldots X_n) \Rightarrow$ joint distribution of $Y = (Y_1 \ldots Y_m)$ • discrete $g(y_1 \ldots y_m) = \sum_{x : r(x) = y} f(x)$
- Continuous CDF: $G(y) = P(Y_1 \leq y_1 \ldots) = \int_{x : r(x) \leq y} f(x)\,dx$
- Convolution method: $Y = X_1 + X_2 \Rightarrow g(y) = \int f_1(z) f_2(y - z)\,dz$

- Transformation method: $n=m$, $x=s(y) \Rightarrow g(y)=f(s(y))\,|J|$

$$J=\det\begin{bmatrix}\frac{ds_1}{dy_1} & \cdots & \frac{ds_1}{dy_n}\\ \frac{ds_n}{dy_1} & & \frac{ds_n}{dy_n}\end{bmatrix}$$

- Bernoulli → two outcomes (success = 1, failure = 0)

  parameter p = probability of success

  $f(x)=p^x(1-p)^{1-x}$

  mean $= p$ variance $= p(1-p)$

- Binomial → total no of successes in n Bernoulli trials

  parameters: n, p mean: np mgf: $\psi(t)$

  $f(x)={}^nC_x\,p^x(1-p)^{n-x}$ variance: $np(1-p)$ $(pe^t+1-p)^n$

  $x_i \sim Bin(n_i,p) \Rightarrow \Sigma x_i \sim Bin(\Sigma n_i, p)$

- poisson → the no of outcomes in fixed intervals of time

  parameter: λ $f(x)=e^{-\lambda}x^x/x!$ mean $=\lambda$ var $=\lambda$

  $\psi(t)=e^{\lambda(e^t-1)}$ $x_i \sim poisson(\lambda_i) \Rightarrow \Sigma x_i \sim poisson(\Sigma\lambda_i)$

  $n \to$ large, $p \to$ small $\Rightarrow Bin(n,p) \approx poisson(np)$

- -ve binomial: no of failures in sequence of Bernoulli trials before r^{th} success.

  parameters: r, p $f(x)=\binom{r+x-1}{x}p^r(1-p)^x$ $x=0,1,2\ldots$

  $E[x]=\frac{r(1-p)}{p}$ $Var=\frac{r(1-p)}{p^2}$ $\psi(t)=\left(\frac{p}{1-(1-p)e^t}\right)^r$

- Geometric: $r=1$ for -ve bin $P(X=k+t/X\geq k)=P(X=t)$

- Normal: (bell-curve)

  parameters: μ, σ^2 $f(x)=\frac{1}{\sqrt{2\pi}\,\sigma}e^{-\left(\frac{(x-\mu)^2}{2\sigma^2}\right)}$ $\psi(t)=e^{\mu t+\sigma^2/2\,t^2}$

  $N(0,1) \to$ standard $\phi(z)=\frac{1}{\sqrt{2\pi}}e^{-z^2/2}$ $\phi(-z)=1-\phi(z)$

  $x_i \sim N(\mu_i,\sigma_i^2) \Rightarrow y=\Sigma a_i x_i+b \sim N(\Sigma a_i\mu_i+b, \Sigma a_i^2\sigma_i^2)$

Standardization: $X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X-\mu}{\sigma} \sim N(0,1)$

$X_1 \dots X_n \sim iid\, N(\mu, \sigma^2) \Rightarrow \bar{X_n}$ (sample mean) $\sim N(\mu, \sigma^2/n)$

- multinomial dist: n trials, k outcomes w p_k $\Sigma p_i = 1$

$$f(x_1 x_2 \dots x_k) = \binom{n}{x_1 x_2 \dots x_k} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k} \qquad E[x_i] = np_i$$

$$Var[x_i] = np_i(1-p_i)$$

- Bivariate normal:

parameters: $\mu_1\ \mu_2\ \sigma_1^2\ \sigma_2^2\ \rho$

$$f(x_1,x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2}{2(1-\rho^2)}\right]$$

$$E[x_2/x_1] = \mu_2 + \frac{\rho\sigma_2}{\sigma_1}(x_1-\mu_1) \qquad Var(x_2/x_1) = \sigma_2^2(1-\rho^2)$$

↓ doesn't depend on x_1

$Y = a_1X_1 + a_2X_2 + b \Rightarrow E[Y] = a_1\mu_1 + a_2\mu_2 + b$ ↳ normal

$$Var(Y) = a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + 2a_1a_2\rho\sigma_1\sigma_2$$